

# Machine Learning-Based Daily Traffic and Toll Revenue Forecasting for the Padma Bridge

Forecasting, explainability and transport-impact analysis

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CSE 4112: Machine Learning Laboratory  
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## Executive summary

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This project was carried out to build a practical one-day-ahead forecasting system for Padma Bridge road traffic and toll revenue. Three datasets were used: daily Padma Bridge records with weather and calendar information, daily Jamuna Bridge records, and district-level nighttime-light observations. After cleaning and date alignment, 1,425 complete daily records and 51 leakage-safe predictors were available for forecasting.

Seven forecasting approaches were compared under the same chronological split. Random Forest produced the lowest traffic MAE and MAPE on the final test period, while the custom Residual MLP produced the lowest toll MAE and MAPE. XGBoost was a close second in both tasks and handled the largest Eid-period errors more consistently. The final traffic MAPE was 8.58%, and the final toll MAPE was 9.47%. These correspond to intuitive MAPE-derived accuracies of 91.42% and 90.53%, although MAPE and R-squared are the proper regression measures.

### Main finding

The models were very reliable on ordinary days, with traffic and toll MAPE values of 4.36% and 3.82% for their respective best average-error models. The difficult cases were public holidays and sharp Eid-related demand changes. The project therefore succeeded as a daily forecasting system, but it also showed exactly where additional data would be most useful.

## Work completed

- Cleaned and audited the supplied bridge and nighttime-light datasets.
- Created lag, rolling, calendar, Eid, weather, railway and cross-bridge features without using future target information.
- Trained baseline, statistical, tree-ensemble and neural-network forecasting models.
- Evaluated every model on a later, untouched 214-day test period.
- Used permutation importance and SHAP to interpret the Random Forest forecasts.
- Performed feature ablation, weather and Eid analysis, Padma-Jamuna dynamic analysis, railway interrupted time-series analysis and nighttime-light Difference-in-Differences.

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# 1. Introduction

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The Padma Bridge carries road traffic between the southwestern region and the rest of Bangladesh. Daily use of the bridge is not constant: weekends, public holidays, Eid travel, weather, seasonal movement and recent traffic conditions all affect demand. A useful forecasting model must therefore do more than learn an average value. It must recognize weekly patterns, recent changes and occasional surges.

The main forecasting targets in this work were total daily vehicle traffic and total daily toll revenue. Both were treated as regression problems. Alongside forecasting, the project examined how weather and events were related to traffic, whether Padma and Jamuna traffic moved together over time, whether commercial railway service was associated with a change in road traffic, and whether nighttime-light growth differed between southwestern and control districts after the bridge opened.

## 1.1 Objectives

- Prepare a reliable daily dataset from the available Padma, Jamuna, weather, holiday and satellite records.
- Forecast next-day Padma traffic and toll revenue using information that would be available at prediction time.
- Compare simple baselines with statistical, ensemble and custom neural models.
- Explain the strongest models and identify which groups of predictors produced the largest improvement.
- Study event effects and broader transport or economic associations without overstating causality.

## 1.2 Scope of the forecasts

The forecast horizon is one day. To predict day  $t$ , the models use calendar variables and expected weather for day  $t$ , together with Padma and Jamuna information observed up to day  $t-1$ . During the rolling test evaluation, the actual completed record from the previous test day becomes available before the next forecast. This is realistic for daily operation, but it is not the same as forecasting several weeks recursively from one starting date.

## 2. Dataset information

Only the three richest non-duplicative CSV files were used. The other supplied files contained subsets or intermediate versions of the same observations; concatenating them would have duplicated dates and produced misleading sample sizes.

| Dataset          | Size                  | Coverage                  | Main information                                                                    |
|------------------|-----------------------|---------------------------|-------------------------------------------------------------------------------------|
| Padma daily      | 1,520 days            | 26 Jun 2022 - 24 Aug 2026 | Traffic by side, toll by side, totals, holiday/Eid flags and six weather variables  |
| Jamuna daily     | 1,515 days            | 1 Jul 2022 - 24 Aug 2026  | East/west traffic and toll values with daily totals                                 |
| Nighttime lights | 5,376 district-months | Jan 2019 - Dec 2025       | 64 districts x 84 months; light-intensity summaries and Padma treatment-group flags |

### 2.1 Padma Bridge data

The Padma file contains daily traffic recorded at Mawa and Jajira, daily cash collection from both sides, published total traffic and total cash, holiday names, holiday and Eid indicators, weekend status, distance to the nearest Eid, mean/maximum/minimum temperature, rainfall, humidity and wind speed. The forecasting targets were the published Total\_Traffic and Total\_Cash fields.

### 2.2 Jamuna Bridge data

The Jamuna file contains daily east-side and west-side vehicle counts and toll collections. Jamuna data were not used as same-day information. Historical Jamuna values were shifted before they entered the Padma forecast so that the model could only use information that would already be known.

### 2.3 Nighttime-light data

The nighttime-light panel covers 64 districts, including 21 southwestern districts and 43 control districts. Monthly mean light intensity was used as a proxy for local economic activity. The panel was used only for the regional before-after analysis; it was not mixed into the daily traffic forecasting table.

Final forecasting sample

After reindexing and removing rows without the required history, 1,425 complete daily rows remained, each with 51 predictors.

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## 3. Data preprocessing and feature engineering

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### 3.1 Numeric cleaning

Traffic and toll columns were originally stored as text because many numbers used Indian comma grouping. Some fields also contained malformed punctuation. A dedicated parser removed grouping commas and recovered valid numeric values from entries such as 1,08,87,150.00. The cleaning step was applied before totals, lags or rolling averages were calculated.

### 3.2 Date audit and alignment

Dates were converted to a common daily format, sorted, checked for duplicates and aligned. The Padma series had no duplicate dates but was missing 31 March 2026. The Jamuna series had no duplicate dates but was missing 17 September 2025. A continuous Padma calendar was created so that a one-row shift could not silently be mistaken for a one-day shift when a date was absent.

| Audit check          | Mismatched dates | Treatment                                  |
|----------------------|------------------|--------------------------------------------|
| Padma traffic total  | 2                | Published total retained; mismatch flagged |
| Padma cash total     | 10               | Published total retained; mismatch flagged |
| Jamuna traffic total | 8                | Published total retained; mismatch flagged |
| Jamuna cash total    | 8                | Published total retained; mismatch flagged |

The mismatches were not silently corrected by adding the side values. The published total remained the target, and every inconsistent date was written to the audit report. This decision keeps the data source traceable and avoids inventing values during preprocessing.

### 3.3 Calendar, event and weather features

- Time index and cyclic sine/cosine encodings for day of week, month and day of year.
- Weekend, public holiday and Eid indicators.
- Distance to the nearest Eid and a signed Eid-relative day, allowing the model to distinguish before-Eid travel from after-Eid travel.
- Temperature, rainfall, humidity and wind speed, together with rainy-day and heavy-rain indicators.
- Rail-service indicator and number of days after regular commercial rail service began.

## 3.4 Historical features and data split

### 3.4 Historical features

| Feature group       | Variables created                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Padma traffic       | Lags 1, 2, 3, 7, 14 and 28; rolling means 7, 14 and 30; rolling standard deviations; short-term change and trend |
| Padma toll          | Cash lags 1, 7 and 14; 7- and 30-day rolling means                                                               |
| Revenue per vehicle | Previous-day value and 7-day rolling mean                                                                        |
| Jamuna traffic      | Lags 1, 2, 7 and 14; 7- and 30-day rolling means                                                                 |
| Jamuna toll         | Cash lags 1 and 7; 7-day rolling mean                                                                            |

### 3.5 Leakage prevention

The most important design rule was that every traffic, toll and Jamuna historical feature had to be shifted by at least one day. Current-day Mawa traffic, Jajira traffic, cash collections and Jamuna totals were excluded from the input. Otherwise the model would indirectly receive the answer it was expected to predict. Rolling means also ended at day  $t-1$ , not day  $t$ .

Forecasting relationship

Known calendar + expected weather + history through day  $t-1$  -> traffic or toll on day  $t$

### 3.6 Chronological split

| Partition  | Rows | Date range                | Use                                                    |
|------------|------|---------------------------|--------------------------------------------------------|
| Training   | 997  | 31 Jul 2022 - 22 Apr 2025 | Model fitting and expanding-window tuning              |
| Validation | 214  | 23 Apr 2025 - 22 Dec 2025 | Comparison of model families and neural stopping epoch |
| Test       | 214  | 23 Dec 2025 - 24 Aug 2026 | One final, untouched evaluation                        |

No random split was used. Random Forest and XGBoost tuning used expanding-window time-series splits. Final configurations were refitted on training plus validation data and evaluated once on the chronological test period.

## 4. Forecasting models

| Model             | Basic idea                                                           | Role in the study                                  |
|-------------------|----------------------------------------------------------------------|----------------------------------------------------|
| Persistence (t-1) | Predict today as yesterday's value                                   | Simple daily benchmark                             |
| Seasonal (t-7)    | Predict today as the same weekday last week                          | Weekly benchmark                                   |
| Linear Regression | Single linear relationship across all predictors                     | Interpretable conventional baseline                |
| SARIMAX           | Autoregressive, moving-average, weekly seasonal and external effects | Statistical time-series baseline                   |
| Random Forest     | Many randomized regression trees combined by averaging               | Nonlinear ensemble; resistant to scaling issues    |
| XGBoost           | Sequentially boosted decision trees                                  | Strong nonlinear tabular model                     |
| Residual MLP      | Two-branch neural network that predicts change from yesterday        | Custom model for historical and external variables |

### 4.1 Random Forest and XGBoost tuning

The full Random Forest search sampled 24 combinations of tree count, maximum features, minimum leaf size, maximum depth and row-sampling fraction. Candidate configurations were ranked by mean chronological validation MAE. XGBoost was tuned in the same time-aware manner over tree depth, learning rate, tree count, row and column sampling, minimum child weight and regularization.

### 4.2 SARIMAX

SARIMAX used recent target history together with calendar, event, weather and railway variables. The selected traffic specification was ARIMA(2,1,1) with a weekly seasonal AR(1) component. The selected toll specification was ARIMA(1,0,1) with the same weekly seasonal component.

### 4.3 Custom Residual MLP

The custom neural model was intentionally small because the dataset contained only about 1,400 complete forecasting rows. One branch processed weather, calendar, Eid and railway features; the second branch processed Padma and Jamuna history. Their hidden representations were joined, and the network predicted a residual rather than the complete target:

Residual prediction

Predicted value on day  $t$  = observed value on day  $t-1$  + network-predicted change

The network used GELU activations, dropout, AdamW optimization, Huber loss and early stopping. Three seeds (1, 7 and 42) were trained and averaged. The final training length was 82 epochs for traffic and 61 epochs for toll revenue.

### 4.4 Evaluation measures

MAE was treated as the main practical measure because it gives the average absolute daily error in vehicles or BDT. RMSE gives more weight to large misses, which is important around Eid. MAPE expresses error as a percentage, while R-squared measures the share of test-set variation explained by the model. For an intuitive percentage only,  $100 - \text{MAPE}$  is reported as MAPE-derived forecasting accuracy; it is not classification accuracy.

## 5. Traffic forecasting results

Random Forest gave the lowest overall traffic MAE and MAPE on the 214-day test period. Its average absolute error was 1,451 vehicles and its MAPE was 8.58%. XGBoost was almost tied in MAE - only about three vehicles worse on average - but it had the lowest RMSE and highest R-squared. This means Random Forest was slightly better for the typical daily error, whereas XGBoost was better at controlling the largest errors.

| Model             | MAE (vehicles) | RMSE  | MAPE   | R-squared | 100 - MAPE |
|-------------------|----------------|-------|--------|-----------|------------|
| Persistence (t-1) | 2,433          | 4,008 | 12.72% | 0.490     | 87.28%     |
| Seasonal (t-7)    | 3,362          | 6,115 | 16.18% | -0.187    | 83.82%     |
| Linear Regression | 2,070          | 3,776 | 11.55% | 0.548     | 88.45%     |
| SARIMAX           | 2,105          | 3,980 | 11.21% | 0.497     | 88.79%     |
| Random Forest     | 1,451          | 2,963 | 8.58%  | 0.721     | 91.42%     |
| XGBoost           | 1,454          | 2,876 | 9.09%  | 0.738     | 90.91%     |
| Residual MLP      | 1,530          | 3,150 | 8.66%  | 0.685     | 91.34%     |

### Traffic model choice

Random Forest is the winner when MAE and MAPE are the priority. XGBoost is the better choice when high-demand days and squared-error performance matter more. Both clearly outperform the persistence and weekly baselines.

Compared with the previous-day persistence baseline, Random Forest reduced traffic MAE by 40.4%. Compared with the seven-day seasonal baseline, the reduction was 56.8%. The negative R-squared of the weekly baseline shows that copying the previous week's value was worse than using the test-period mean during this particular period.

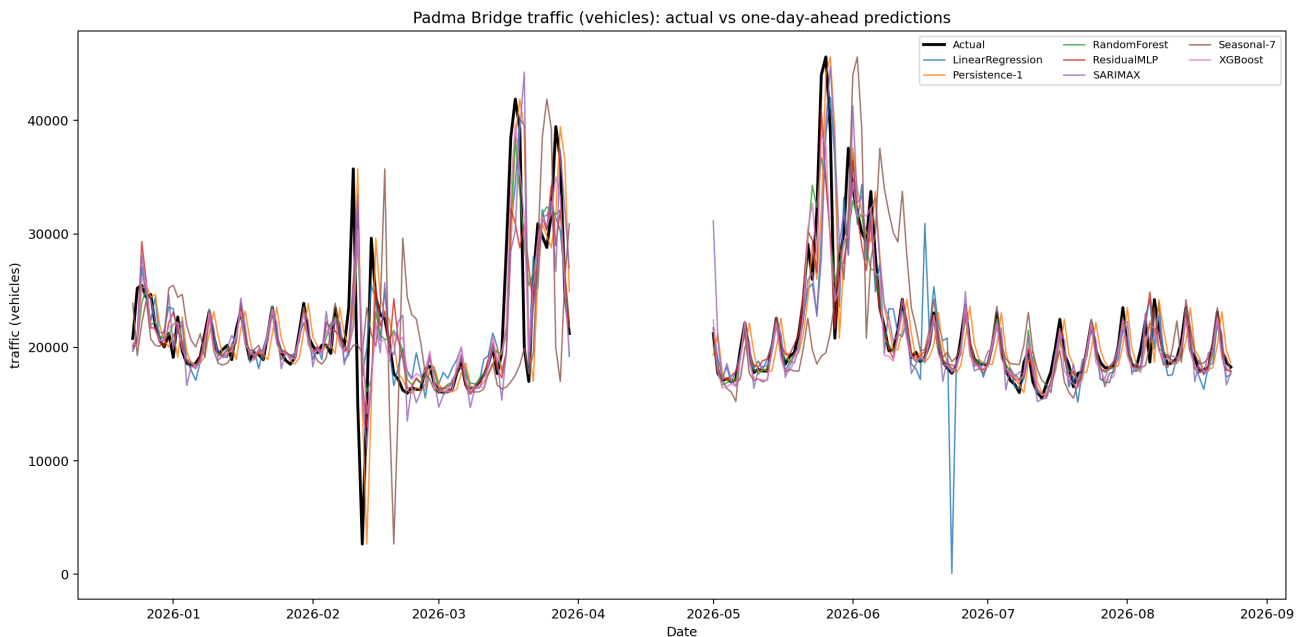


Figure 1. Actual and one-day-ahead traffic predictions during the final test period. Gaps reflect dates for which complete model inputs were unavailable.

## 6. Toll-revenue forecasting results

The Residual MLP obtained the lowest toll MAE and MAPE. Its average absolute error was about BDT 1.519 million and its MAPE was 9.47%. XGBoost was again very close and produced the lowest RMSE and highest R-squared. The difference in MAE between the two models was only about BDT 25,000, or 1.6% of the Residual MLP's MAE.

| Model             | MAE (million BDT) | RMSE (million) | MAPE   | R-squared | 100 - MAPE |
|-------------------|-------------------|----------------|--------|-----------|------------|
| Persistence (t-1) | 2.320             | 4.116          | 12.38% | 0.390     | 87.62%     |
| Seasonal (t-7)    | 3.517             | 6.184          | 15.90% | -0.376    | 84.10%     |
| Linear Regression | 2.119             | 3.789          | 11.49% | 0.483     | 88.51%     |
| SARIMAX           | 2.141             | 3.647          | 11.55% | 0.521     | 88.45%     |
| Random Forest     | 1.626             | 3.040          | 9.75%  | 0.667     | 90.25%     |
| XGBoost           | 1.544             | 2.918          | 9.56%  | 0.693     | 90.44%     |
| Residual MLP      | 1.519             | 2.997          | 9.47%  | 0.677     | 90.53%     |

### Toll model choice

The custom Residual MLP is the best average-error model. XGBoost is the most balanced model because it has slightly better RMSE, R-squared and peak-period performance.

The Residual MLP reduced toll MAE by 34.5% relative to the previous-day persistence baseline and by 56.8% relative to the seven-day seasonal baseline. Its success supports the residual design: toll revenue usually follows recent levels, and the network only needs to learn the adjustment associated with changing traffic, events and external conditions.

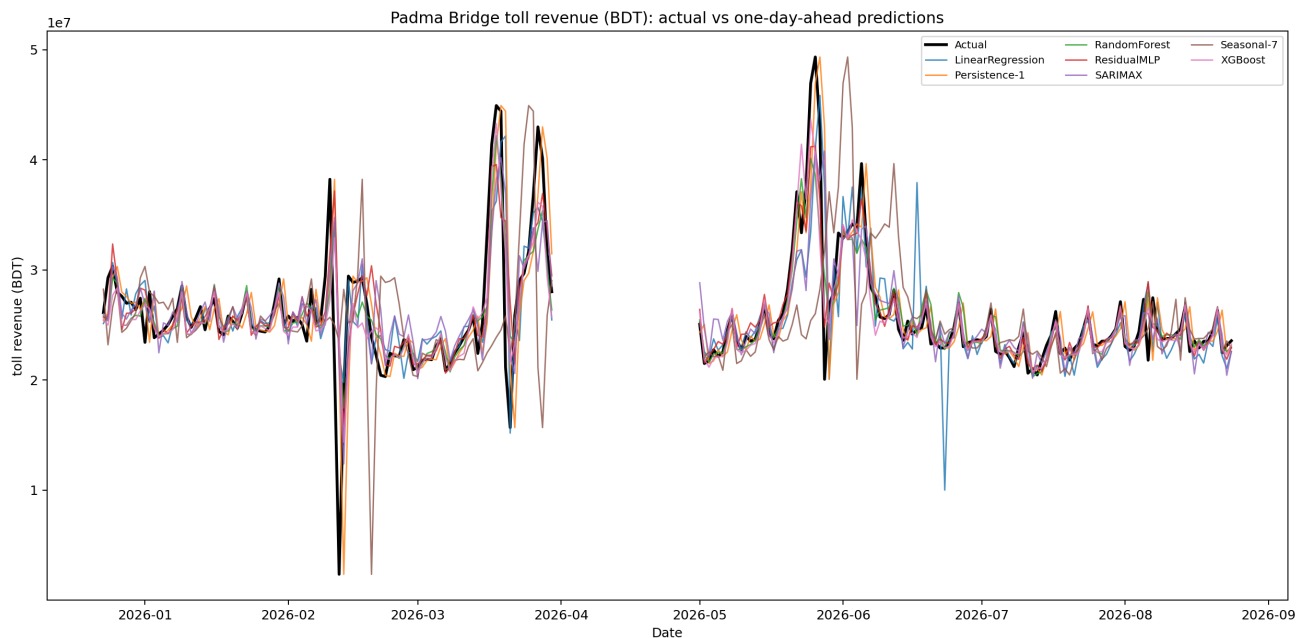


Figure 2. Actual and one-day-ahead toll-revenue predictions during the final test period.

## 7. Performance on normal days, holidays and Eid periods

Overall averages can hide operational weaknesses. The test results were therefore recalculated for normal days, public holidays and dates within seven days of Eid. The table below compares the leading models for each target.

| Target  | Model         | Segment        | MAE*  | MAPE   | R-squared |
|---------|---------------|----------------|-------|--------|-----------|
| Traffic | Random Forest | Normal         | 996   | 4.36%  | 0.783     |
| Traffic | Random Forest | Public holiday | 5,057 | 42.04% | 0.540     |
| Traffic | Random Forest | Eid +/- 7 days | 3,957 | 13.25% | 0.521     |
| Traffic | XGBoost       | Normal         | 1,052 | 4.70%  | 0.800     |
| Traffic | XGBoost       | Public holiday | 4,638 | 43.86% | 0.562     |
| Traffic | XGBoost       | Eid +/- 7 days | 3,373 | 11.18% | 0.645     |
| Toll    | Residual MLP  | Normal         | 1.017 | 3.82%  | 0.805     |
| Toll    | Residual MLP  | Public holiday | 5.492 | 54.21% | 0.503     |
| Toll    | Residual MLP  | Eid +/- 7 days | 3.875 | 12.97% | 0.599     |
| Toll    | XGBoost       | Normal         | 1.121 | 4.12%  | 0.755     |
| Toll    | XGBoost       | Public holiday | 4.894 | 52.63% | 0.591     |
| Toll    | XGBoost       | Eid +/- 7 days | 3.569 | 11.51% | 0.672     |

\*Traffic MAE is in vehicles; toll MAE is in million BDT.

Normal-day results were strong. Random Forest traffic MAPE fell to 4.36%, and Residual MLP toll MAPE fell to 3.82%. Holiday performance was much weaker: the corresponding MAPEs were 42.04% and 54.21%. Only 24 public holidays were present in the test period, and those days included several unusual demand levels, so the models had much less repeated evidence from which to learn.

XGBoost handled the Eid window better than the average-error winners. Its traffic Eid-window MAE was 3,373 vehicles compared with 3,957 for Random Forest. Its toll Eid-window MAE was BDT 3.569 million compared with BDT 3.875 million for the Residual MLP. This is why the final model decision depends on whether routine accuracy or surge robustness is more important.

### Interpretation

The project achieved strong generalization for routine operation. The remaining problem is not ordinary-day prediction; it is forecasting rare event timing and the size of sudden travel surges.

## 8. Model explainability

Random Forest was explained using three complementary methods: built-in impurity importance, test-set permutation importance and SHAP values. Permutation importance was emphasized because it measures how much test MAE increases when one predictor is shuffled. SHAP was used to examine the direction and size of contributions for individual observations.

| Rank | Traffic feature           | MAE increase | Toll feature                    | MAE increase (BDT) |
|------|---------------------------|--------------|---------------------------------|--------------------|
| 1    | days_to_nearest_eid       | 575          | jamuna_traffic_lag_1            | 679,214            |
| 2    | padma_traffic_lag_1       | 521          | days_to_nearest_eid             | 408,395            |
| 3    | dow_cos                   | 311          | padma_cash_lag_1                | 233,082            |
| 4    | jamuna_traffic_lag_1      | 166          | jamuna_cash_lag_1               | 183,209            |
| 5    | padma_cash_lag_1          | 154          | dow_cos                         | 128,962            |
| 6    | padma_cash_roll_mean_7    | 93           | padma_cash_lag_7                | 118,971            |
| 7    | revenue_per_vehicle_lag_1 | 81           | padma_cash_roll_mean_7          | 91,156             |
| 8    | padma_roll7_minus_roll30  | 79           | revenue_per_vehicle_roll_mean_7 | 63,195             |

For traffic, the two dominant predictors were distance to Eid and the previous day's Padma traffic. The day-of-week cycle, previous Jamuna traffic and previous Padma toll were also important. For toll revenue, previous Jamuna traffic had the largest permutation effect, followed by distance to Eid and previous Padma cash. These rankings agree with the idea that bridge demand is driven mainly by recent mobility and event timing rather than weather alone.

The high importance of Jamuna history should be interpreted carefully. It shows that Jamuna records contain useful predictive information, but it does not prove that traffic on one bridge directly causes traffic on the other. Shared holidays, national travel demand and reporting patterns can affect both series.

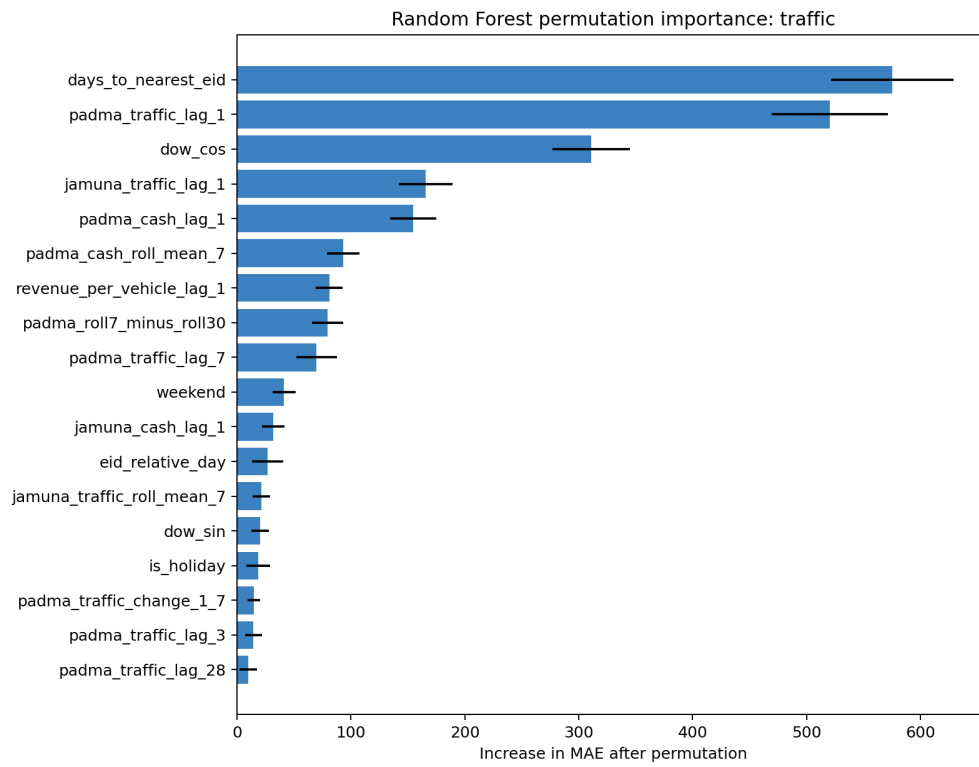


Figure 3. Traffic permutation importance. A longer bar means a larger rise in MAE when that feature is disrupted.

## 9. Feature-ablation study

The ablation study trained the same Random Forest with progressively richer feature groups. This made it possible to see whether the final performance came from weather, calendar information or recent bridge history.

| Feature set           | No. | Traffic MAE | Traffic MAPE | Traffic R2 | Toll MAE (m) | Toll MAPE | Toll R2 |
|-----------------------|-----|-------------|--------------|------------|--------------|-----------|---------|
| Weather Only          | 8   | 4,230       | 19.20%       | -0.469     | 3.836        | 16.89%    | -0.420  |
| Weather Calendar      | 22  | 2,238       | 12.98%       | 0.555      | 2.240        | 12.88%    | 0.478   |
| Add Padma History     | 35  | 1,493       | 8.58%        | 0.734      | 1.692        | 10.15%    | 0.660   |
| Add Jamuna History    | 44  | 1,404       | 8.18%        | 0.744      | 1.663        | 10.11%    | 0.663   |
| Full Add Toll History | 51  | 1,421       | 8.22%        | 0.738      | 1.628        | 9.79%     | 0.674   |

Weather alone was insufficient for either task and produced negative R-squared values. Adding calendar and event variables gave a large improvement, but the decisive change came when Padma traffic history was included. Traffic MAPE decreased from 12.98% to 8.58%, and toll MAPE decreased from 12.88% to 10.15%.

Jamuna history improved traffic MAE from 1,493 to 1,404 vehicles and reduced traffic MAPE to 8.18%. Adding toll-history features after that gave a very small deterioration in traffic performance, suggesting some redundant information. For toll forecasting, however, the complete feature set was best: toll MAPE reached 9.79%. This is sensible because past cash collection and revenue per vehicle are directly relevant to future toll revenue.

What the ablation shows

Feature engineering mattered more than simply choosing a more complicated model. Recent Padma history created the largest gain, Jamuna history added useful cross-corridor information, and toll history was especially valuable for the toll target.

## 10. Weather, weekend, holiday and Eid analysis

| Condition      | Days | Mean traffic | Descriptive difference |
|----------------|------|--------------|------------------------|
| Weekday        | 1086 | 17,595       | Reference              |
| Weekend        | 434  | 20,288       | +15.3% vs weekday      |
| Non-holiday    | 1405 | 17,965       | Reference              |
| Public holiday | 115  | 23,237       | +29.3% vs non-holiday  |
| Non-Eid        | 1480 | 18,076       | Reference              |
| Eid            | 40   | 29,023       | +60.6% vs non-Eid      |
| No heavy rain  | 1427 | 18,413       | Reference              |
| Heavy rain     | 93   | 17,613       | -4.3% vs no heavy rain |

The largest descriptive differences were connected to the travel calendar. Weekend traffic was 15.3% higher than weekday traffic, public-holiday traffic was 29.3% higher than non-holiday traffic, and the mean on Eid-labelled days was 60.6% higher than on other days. Heavy-rain days had 4.3% lower mean traffic. Spearman correlations between traffic and individual weather variables were weak: the largest absolute value was only 0.080 for wind speed.

The Eid profile gives a more detailed picture than a single binary flag. Mean traffic rose sharply three days before Eid and peaked two days before Eid at about 38,602 vehicles. Traffic fell on Eid day itself, then rose again after Eid and reached another local peak around six days later. This explains why both distance to Eid and signed Eid-relative day were useful features.

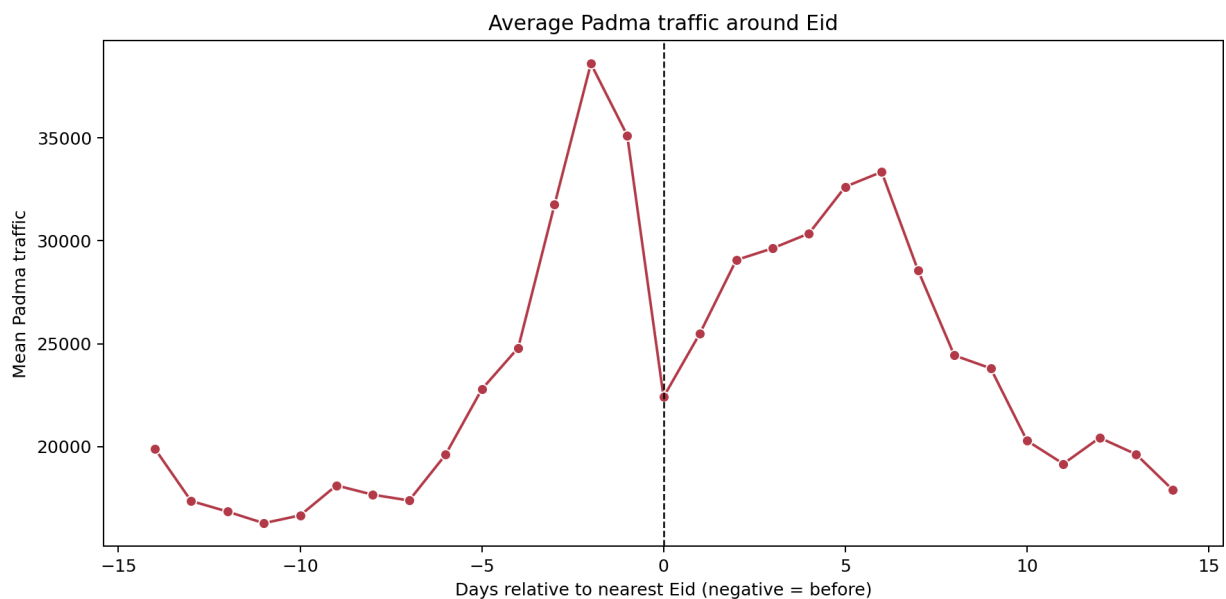


Figure 4. Mean Padma traffic from 14 days before to 14 days after the nearest Eid. Each relative day represents approximately nine Eid observations across the dataset.

## 11. Padma-Jamuna dynamic analysis

Padma and Jamuna traffic both contain trend, weekly seasonality and event-driven spikes. To reduce the chance of finding a misleading relationship simply because both series move over time, the analysis used log-transformed first differences. Augmented Dickey-Fuller tests strongly rejected a unit root for the differenced series.

The contemporaneous correlation between daily log traffic changes was 0.839, indicating that the bridges often experienced demand changes in the same direction on the same day. Positive correlations also appeared at weekly lags. Bidirectional Granger tests were significant: the smallest p-value for Jamuna history predicting Padma changes was  $1.06 \times 10^{-12}$ , and the reverse direction had a minimum p-value of  $1.29 \times 10^{-12}$ . A 14-lag VAR was selected by AIC and used for the impulse-response analysis.

| Past values of | Forecasted series | Best lag | F statistic | p-value  |
|----------------|-------------------|----------|-------------|----------|
| Jamuna         | Padma             | 13       | 6.74        | 1.06e-12 |
| Padma          | Jamuna            | 7        | 10.26       | 1.29e-12 |

These tests establish predictive temporal dependence, not physical causation. Shared national travel demand, weekly schedules, festivals and other common shocks may contribute to both directions. The result nevertheless supports the modeling decision to include lagged Jamuna information in the Padma forecasting table.

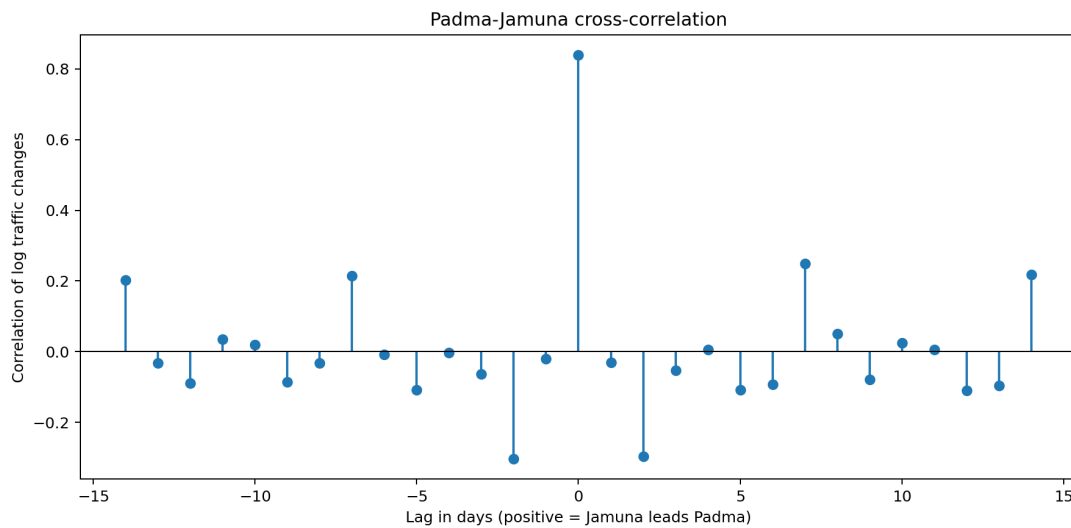


Figure 5. Cross-correlation of daily log traffic changes. Positive lag means Jamuna is earlier than Padma.

## 12. Railway interrupted time-series analysis

Regular commercial passenger rail service through the Padma Bridge was represented as starting on 1 November 2023. The interrupted time-series model separated an immediate level change from a change in the post-opening trend. It also controlled for weekends, public holidays, Eid, rainfall, temperature, humidity, wind, day-of-week and annual seasonality. Heteroskedasticity and autocorrelation-consistent standard errors were calculated with a seven-day lag window.

| Term                   | Estimate         | p-value | Interpretation                                 |
|------------------------|------------------|---------|------------------------------------------------|
| Immediate level change | -3.37%           | 0.460   | Not statistically significant                  |
| Additional daily trend | -0.0367% per day | 0.0052  | Statistically significant negative association |

The estimated immediate change was -3.37%, but its p-value was 0.460, so there was no clear evidence of an abrupt road-traffic drop at the start of commercial rail service. The estimated post-rail trend change was -0.0367% per day and was statistically significant. A sensitivity model that included the previous day's log traffic also found a smaller but significant negative trend change.

This result must remain an association. There was no separate control road corridor and no daily rail-passenger series in the model. Other developments after November 2023 could also change the traffic trend. The railway analysis is therefore useful as an adjusted temporal study, but it is not enough to claim that rail service caused the entire estimated change.

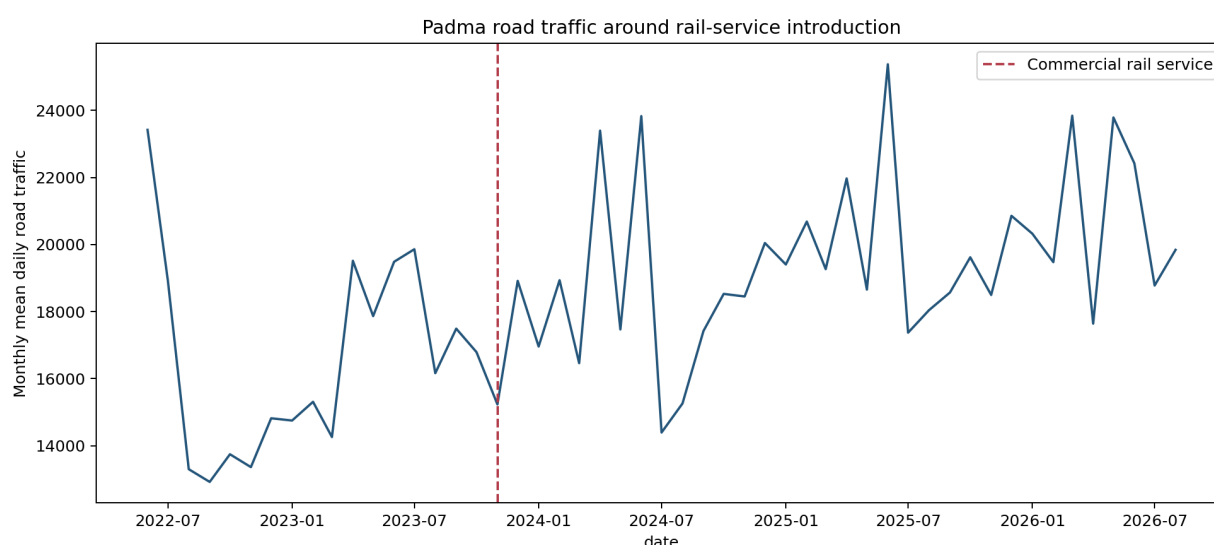


Figure 6. Monthly mean road traffic around the introduction of regular commercial rail service.

## 13. Nighttime-light Difference-in-Differences analysis

Monthly nighttime-light intensity was used as an indirect measure of regional economic activity. The treatment group contained 21 southwestern districts, while the remaining 43 districts formed the comparison group. The main Difference-in-Differences regression included district fixed effects, calendar-month fixed effects and district-clustered standard errors.

| Group                  | Period | Mean NTL | Observations          |
|------------------------|--------|----------|-----------------------|
| Control districts      | Before | 0.602    | 1,806 district-months |
| Control districts      | After  | 0.995    | 1,806 district-months |
| Southwestern districts | Before | 0.208    | 882 district-months   |
| Southwestern districts | After  | 0.449    | 882 district-months   |

Both groups became brighter after the bridge opened, but the increase was smaller in the southwestern group. The estimated DiD coefficient was -0.151 with a district-clustered standard error of 0.071, a p-value of 0.034, and a 95% confidence interval from -0.290 to -0.012.

### Critical limitation

The pre-trend interaction was statistically significant ( $p = 0.0098$ ). Therefore, the parallel-trends assumption is doubtful. The negative DiD estimate should be presented as a descriptive adjusted difference, not evidence that the Padma Bridge reduced economic activity in southwestern districts.

Nighttime lights are also an imperfect proxy. Differences in electrification, cloud correction, land use, urban concentration and satellite measurement can affect observed intensity. A stronger impact study would combine nighttime lights with district income, business registration, employment or transport-flow indicators and would test alternative control groups.

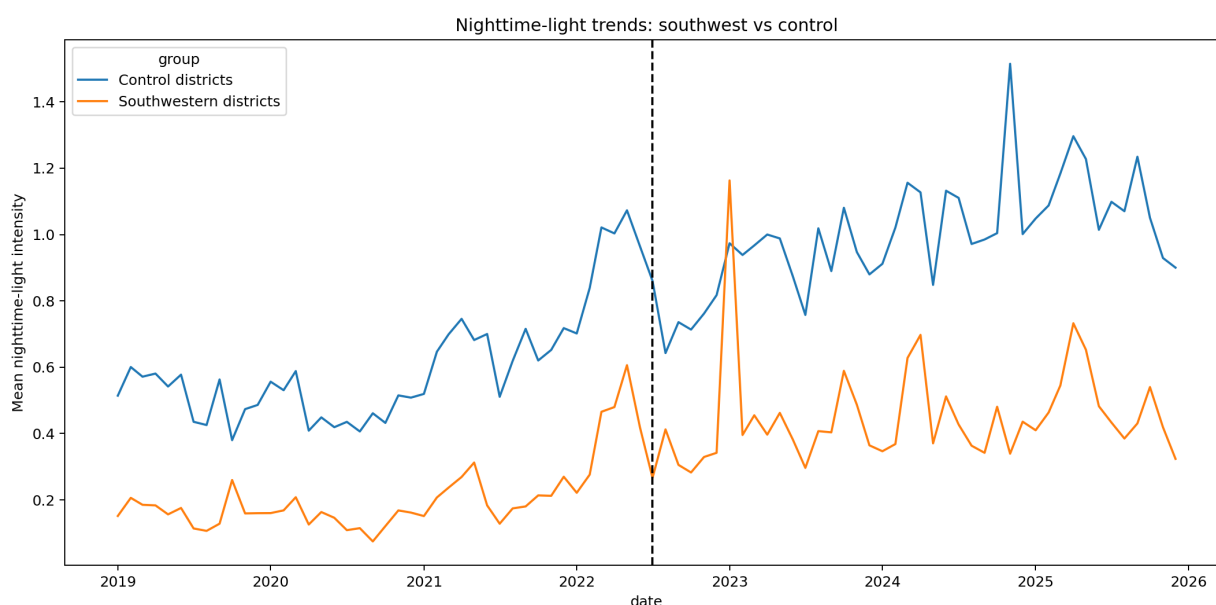


Figure 7. Mean nighttime-light intensity for southwestern and control districts. The vertical line marks the first full post-opening month.

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## 14. Discussion

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### 14.1 Why the forecasting results are strong

The dataset was modest in size, yet the strongest models achieved test MAPEs below 10%. This result is credible because the split followed time order and the final 214 days were not used for hyperparameter selection. The models also beat both simple historical baselines by a wide margin. Most of the gain came from constructing useful, leakage-safe history rather than increasing model size.

Random Forest, XGBoost and the Residual MLP performed much better than Linear Regression and SARIMAX during the final period. This indicates that the relationships among recent traffic, event timing and toll values were nonlinear. It also suggests that a single fixed linear effect for Eid, rainfall or weekend status cannot capture all combinations of conditions.

### 14.2 Why the best model depends on the criterion

No model dominated every metric. Random Forest minimized ordinary absolute error for traffic, but XGBoost better controlled large misses. The Residual MLP minimized toll MAE, but XGBoost had a better toll RMSE and R-squared. For routine daily planning, the MAE winners are reasonable. For capacity planning around Eid, XGBoost deserves priority or could be combined with an event-specific adjustment model.

### 14.3 Main limitations

- Only about four years of daily bridge history were available, giving relatively few Eid and public-holiday examples.
- Weather values are observed historical weather. A live forecast must use weather forecasts, which introduce additional uncertainty.
- Several dates contained inconsistent published side totals, and two daily dates were missing across the bridge series.
- The evaluation is one-day-ahead with newly observed lag values; it does not establish long-horizon recursive accuracy.
- Prediction intervals were not produced, so the point forecasts do not directly communicate uncertainty.
- Granger results indicate predictive dependence rather than physical causation.
- The railway study lacks a control corridor, and the nighttime-light DiD fails a clean parallel-trends check.

### 14.4 Practical next steps

- Add vehicle-class counts, toll-rate revisions, school vacations and verified festival calendars.
- Develop an event-aware model or oversampling strategy specifically for Eid and public holidays.
- Produce prediction intervals using quantile boosting or conformal prediction.
- Retrain periodically as new daily observations arrive and monitor error drift by month and event type.
- For impact analysis, add stronger controls and socioeconomic outcomes before making causal claims.

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## 15. Conclusion

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This project produced a complete and reproducible workflow for Padma Bridge traffic and toll forecasting. The work began with raw records that contained formatted numeric strings, missing dates and inconsistent totals. These records were audited, aligned and converted into a leakage-safe daily modeling table with 51 predictors. Seven forecasting methods were then evaluated using a chronological development and test design.

Random Forest was the strongest average-error traffic model, reaching an MAE of 1,451 vehicles, a MAPE of 8.58% and an R-squared of 0.721. The custom Residual MLP was the strongest average-error toll model, reaching an MAE of BDT 1.519 million, a MAPE of 9.47% and an R-squared of 0.677. XGBoost was nearly tied and gave better RMSE and event-period robustness for both targets.

The detailed analyses showed that recent bridge history and Eid timing were the central predictive signals. Weather had only a weak direct relationship with traffic, while weekends, holidays and Eid were associated with clear increases. Padma and Jamuna traffic changes displayed strong contemporaneous and lagged dependence. The railway and nighttime-light analyses produced statistically interesting associations, but their design limitations prevent strong causal conclusions.

### Overall assessment

Given the available data, the forecasting results are strong. The system performs particularly well on normal days and provides a defensible basis for daily operational forecasting. Its main remaining challenge is accurate prediction during rare holidays and extreme travel periods.

### 15.1 Reproducibility

All randomized components used fixed seeds. The complete run saved processed data, tuning tables, validation and test predictions, metric tables, explainability outputs, statistical summaries and trained model files. The main pipeline can be rerun from one command, while the prediction script uses the saved models for the day immediately following the latest observed record.

## Appendix A. Selected final configurations

| Model                 | Selected configuration                                                            |
|-----------------------|-----------------------------------------------------------------------------------|
| Traffic Random Forest | 1,000 trees; max_features 0.35; leaf size 1; depth 15; full row sample            |
| Toll Random Forest    | 800 trees; max_features 0.75; leaf size 4; unlimited depth; full row sample       |
| Traffic XGBoost       | 500 trees; depth 2; learning rate 0.03; subsample 0.85; colsample 0.60; lambda 3  |
| Toll XGBoost          | 300 trees; depth 5; learning rate 0.015; subsample 0.85; colsample 0.80; lambda 1 |
| Traffic SARIMAX       | ARIMA(2,1,1) with seasonal (1,0,0,7) and external variables                       |
| Toll SARIMAX          | ARIMA(1,0,1) with seasonal (1,0,0,7) and external variables                       |
| Residual MLP          | External branch 32-16; history branch 64-32; joined 32-16; three-seed ensemble    |

## Appendix B. Files produced

- traffic\_test\_metrics.csv and toll\_test\_metrics.csv - final comparison tables
- traffic\_test\_predictions.csv and toll\_test\_predictions.csv - actual and predicted values
- traffic\_feature\_ablation.csv and toll\_feature\_ablation.csv - feature-group experiments
- Random Forest permutation-importance and SHAP outputs - forecast explanation
- Padma-Jamuna ADF, cross-correlation, Granger, VAR and impulse-response outputs
- Railway interrupted time-series coefficients and summary
- Nighttime-light before-after table, DiD summary and pre-trend test
- Saved Linear Regression, Random Forest, XGBoost, SARIMAX and Residual MLP model files

## References and data sources

1. Bangladesh Bridge Authority. Padma and Jamuna bridge traffic and toll records. <https://bba.gov.bd/>
2. Open-Meteo. Historical weather API documentation. <https://open-meteo.com/en/docs/historical-weather-api>
3. Padma Bridge Rail Link Project. Project information. <https://pbrlp.gov.bd/>
4. World Bank. Space2Stats monthly and annual Black Marble nighttime lights. <https://datacatalog.worldbank.org/search/dataset/0066940/space2stats-monthly-annual-black-marble-nighttime-lights>
5. CSE 4112 Group B2 project brief: Machine Learning-Based Daily Traffic and Toll Revenue Forecasting for the Padma Bridge, 2026.